

NAVIGATION AND COMMUNICATION AT SEA

Specialized tools were used at sea to communicate signals, calculate position, and plot navigational routes. Aside from the magnetic compass, the navigator's principal tools were instruments such as the astrolabe and sextant, used for measuring the altitude of the Pole Star or the sun at noon. In conjunction with astronomical almanacs, these readings indicated the ship's latitude (how far north or south of the equator it was). Until the mid-18th century, longitude (a ship's east–west position) was calculated by “dead reckoning”, using the ship's direction of travel and its speed. However, the invention of timepieces of unprecedented accuracy, such as John Harrison's marine chronometer of 1730–35, and his H4 timepiece of 1759, made precise calculations of longitude possible. The difference between the time on the clock – set to the time at Greenwich in London, England – and noon – as observed from the sun's position – indicated how many degrees the ship had sailed east or west of Greenwich.



▲ BUTTERFIELD DIAL
Date Late 17th century
Origin France
Material Silver, glass

A Butterfield dial was a small portable sundial with an incorporated compass. The dials took their name from their inventor, Michael Butterfield (1635–1724), an English instrument-maker who worked in Paris. The gnomon (the raised blade that casts the shadow) was adjusted according to the latitude.

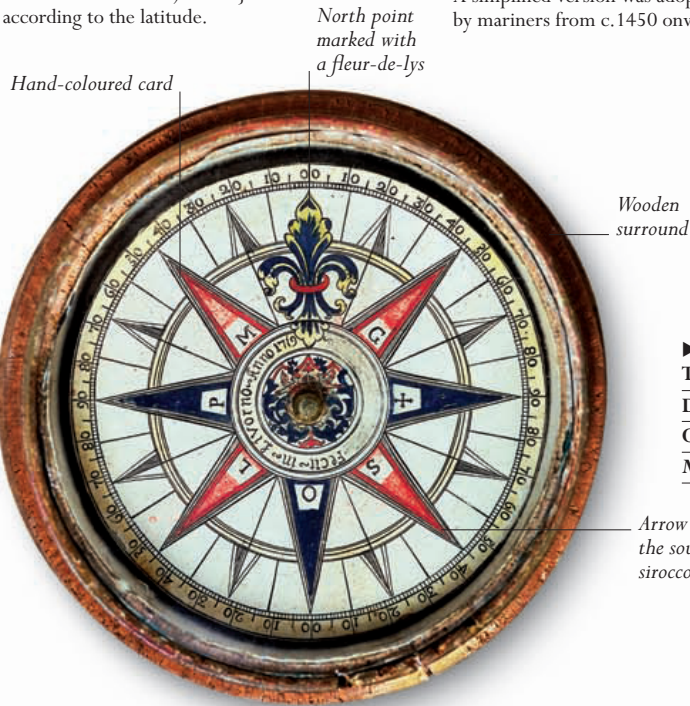


▲ ASTROLABE
Date 1690
Origin Austria
Material Brass

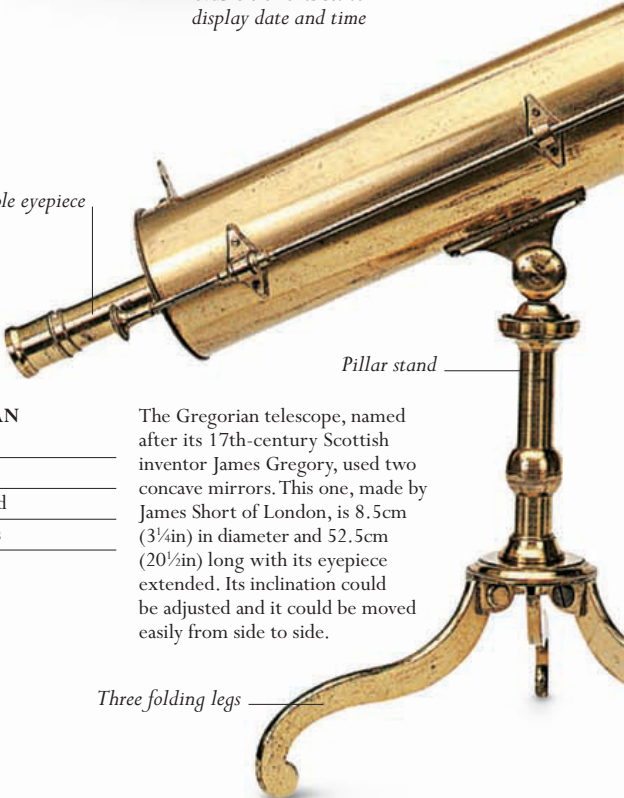
The astrolabe was used by medieval astronomers to observe the stars. A simplified version was adopted by mariners from c.1450 onwards.

► NAVIGATIONAL COMPASS
Date c.1719
Origin Italy
Material Wood, card

Developed in China, the navigational compass came to be used in Europe by the 12th century. On this mariner's compass, a quadrantal degree scale runs around the outer edge of the face. The cardinal and intercardinal points are identified by the initial letter of the Italian name for the wind that blows from that direction. North is the exception, being marked by a fleur-de-lys.



► GREGORIAN TELESCOPE
Date 1752
Origin England
Material Brass



The Gregorian telescope, named after its 17th-century Scottish inventor James Gregory, used two concave mirrors. This one, made by James Short of London, is 8.5cm (3¼in) in diameter and 52.5cm (20½in) long with its eyepiece extended. Its inclination could be adjusted and it could be moved easily from side to side.